Major-Project Proposal on

AI Interpretable Deep Learning Framework for Lung Cancer Classification and Nodule Segmentation

ABSTRACT

Lung cancer is one of the leading causes of cancer-related deaths worldwide, making early and accurate detection essential for improving treatment outcomes and patient survival rates. Recent advances in Artificial Intelligence (AI) and Deep Learning have demonstrated significant potential in medical image analysis by enabling automated detection, localization, and classification of lung nodules.

This project presents an **AI Interpretable Deep Learning Framework for Lung Cancer Classification and Nodule Segmentation** that utilizes deep learning techniques for analyzing lung medical images. The proposed framework performs image preprocessing to improve image quality, followed by lung and nodule region analysis for identifying suspicious abnormalities. A deep learning-based classification model is used to classify detected lung nodules based on their learned visual features. To improve the interpretability of the model, **Explainable AI (XAI)** techniques such as **Grad-CAM** are incorporated to generate visual heatmaps highlighting the regions that contribute to the model's prediction.

The proposed system is evaluated using publicly available lung cancer medical imaging datasets and performance is measured using evaluation metrics such as **accuracy, precision, recall, F1-score, and ROC-AUC**. The framework can be integrated into a web-based application to provide an accessible interface for uploading medical images and obtaining model predictions along with visual explanations.

By combining **Deep Learning, Computer Vision, Medical Image Processing, and Explainable AI**, **Edge AI** the proposed framework aims to provide an efficient and interpretable computer-aided system for lung nodule analysis and lung cancer classification. The system is intended as a **research and clinical decision-support prototype** and is not a replacement for professional medical diagnosis.

Keywords: *Lung Cancer Detection, Deep Learning, Artificial Intelligence (AI), Medical Image Analysis, Computer Vision, Lung Nodule Classification, Image Processing, Convolutional Neural Networks (CNN), Vision Transformer (ViT), Edge AI, Explainable AI (XAI), Grad-CAM, Medical Imaging, Tumor Detection, Lung Nodule Detection, Image Segmentation, Computer-Aided Diagnosis, Clinical Decision Support.*

1. **Domain / Research Area:** AI&ML-Health Care/ Medical Image Analysis

   Base Paper Title: "Encouraging Discriminative Attention Through Contrastive Explainability Learning for Lung Cancer Diagnosis" V. Shravya, Meghana Sunil, B. Natarajan, and R. Elakkiya. IEEE Access, 202510.1109/ACCESS.2025.3616056.

2. **System Requirements:**
   a. **Software Requirements:**
      - **Programming Language:** Python 3.11+
      - **Frameworks/Libraries:** PyTorch, TensorFlow, OpenCV, NumPy, Pandas, Scikit-learn, Matplotlib
      - **Deep Learning / AI:** Vision Transformer (ViT), Convolutional Neural Networks (CNN), Grad-CAM
      - **Development Tools:** Jupyter Notebook, Google Colab, Visual Studio Code (VS Code), GitHub
      - **Dataset Sources:** LIDC-IDRI, LUNA16, Kaggle
      - **Web Framework:** Flask / Streamlit
      - **Operating System:** Windows 10/11, Ubuntu 22.04, or macOS
   b. **Hardware Requirements:**
      - **Processor:** Intel Core i5 (10th Gen) / AMD Ryzen 5 or higher
      - **Storage:** 256 GB SSD minimum, 512 GB SSD recommended
      - **RAM:** 8 GB minimum, 16 GB recommended
      - **GPU:** NVIDIA GPU with CUDA support recommended for model training; Google Colab GPU can be used as an alternative.

3. **Particular of Student/(s)**:

S. No.	Name	Roll No.	Gender (M/F)	CGPA (Up to previous Semester)
1.	Polaka Vikram Simha Reddy	23091A05T5	M	8.55
2.	K Veda Sai Bharadwaj	23091A05Q8	M	8.22
3.	Karnati Venkata Sai Bhuvan	23091A05R7	M	6.0
4.	Kalluri Venkata Siva Prasad Reddy	23091A05T0	M	7.16

4. **Project Duration (in months) :** 3 Months (12 Weeks)

5. **Work Plan (including detailed methodology and time schedule):**

Phase	Duration	Activities
Analysis	1 Week	Literature survey, study of Deep Learning and Vision Transformer (ViT) models, dataset selection (LIDC-IDRI, LUNA16, Kaggle), problem definition, and requirement analysis.

Designing	1 Week	Design system architecture, CT image preprocessing pipeline, lung/nodule detection workflow, deep learning classification model, and Explainable AI framework.
Implementation	6 Weeks	**Module 1:** Medical image acquisition and CT image preprocessing. **Module 2**: Lung/nodule detection and segmentation. **Module 3**: Deep Learning/ViT-based feature extraction and nodule classification. **Module 4**: Explainable AI using Grad-CAM/attention heatmaps for model interpretability. **Module 5**: Web application development for image upload, prediction, visualization, and result presentation.
Testing	2 Weeks	Unit and integration testing, model validation, performance evaluation using Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix, and analysis of false-positive and false-negative cases.
Paper Writing	1 Week	Research paper drafting based on CRS validation results and experimental findings.
Publication	1 Week	Submission to IEEE conference or Scopus-indexed journal.

6. Declaration by student/s:

I hereby declare that the project work entitled **"AI Interpretable Deep Learning Framework for Lung Cancer Classification and Nodule Segmentation "** shall be undertaken strictly in accordance with the provisions specified in the project proposal under the guidance and consultation of the project supervisor. I further undertake to submit periodic progress reports and the final project report in the prescribed format of the Department of Computer Science & Engineering, Rajeev Gandhi Memorial College of Engineering and Technology (RGMCET), upon successful completion of the project.

Date:

Place: Nandyal, Andhra Pradesh (Signature of Student/s with Roll No.)

Under the Guidance of:

Dr. N. Madhusudhana Reddy Ph. D
Professor
Computer Science & Engineering, RGMCET

Signature of Guide	**Signature of Co-ordinator**	**Signature of HOD**
(Dr. N. MADHUSUDHANA REDDY)	(Mr. P. NAVEEN SUNDAR KUMAR)	(Dr. K SUBBA REDDY)